

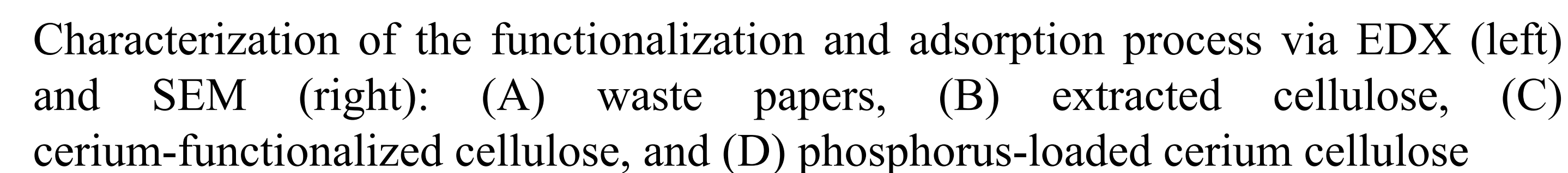
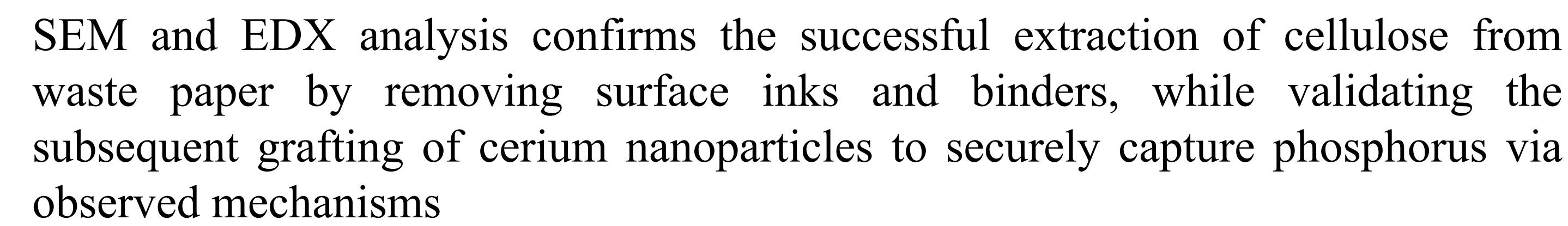
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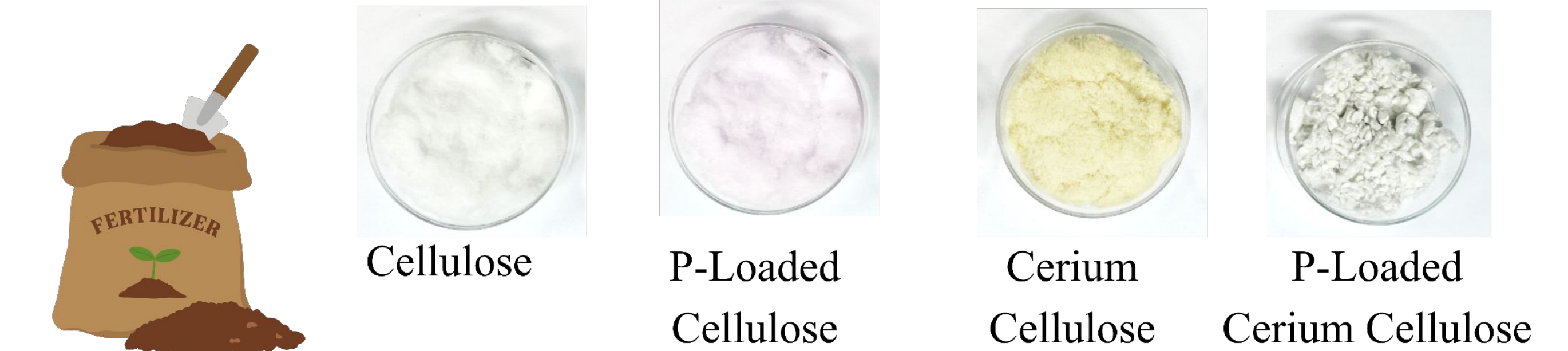


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Testing the Growth: Agricultural Evaluation of Recovered Fertilizers



Agricultural evaluation experiments are actively investigating across cellulose, P-loaded cellulose, cerium cellulose, and P-loaded cerium cellulose to evaluate the adsorbent as a slow-releasing fertilizer

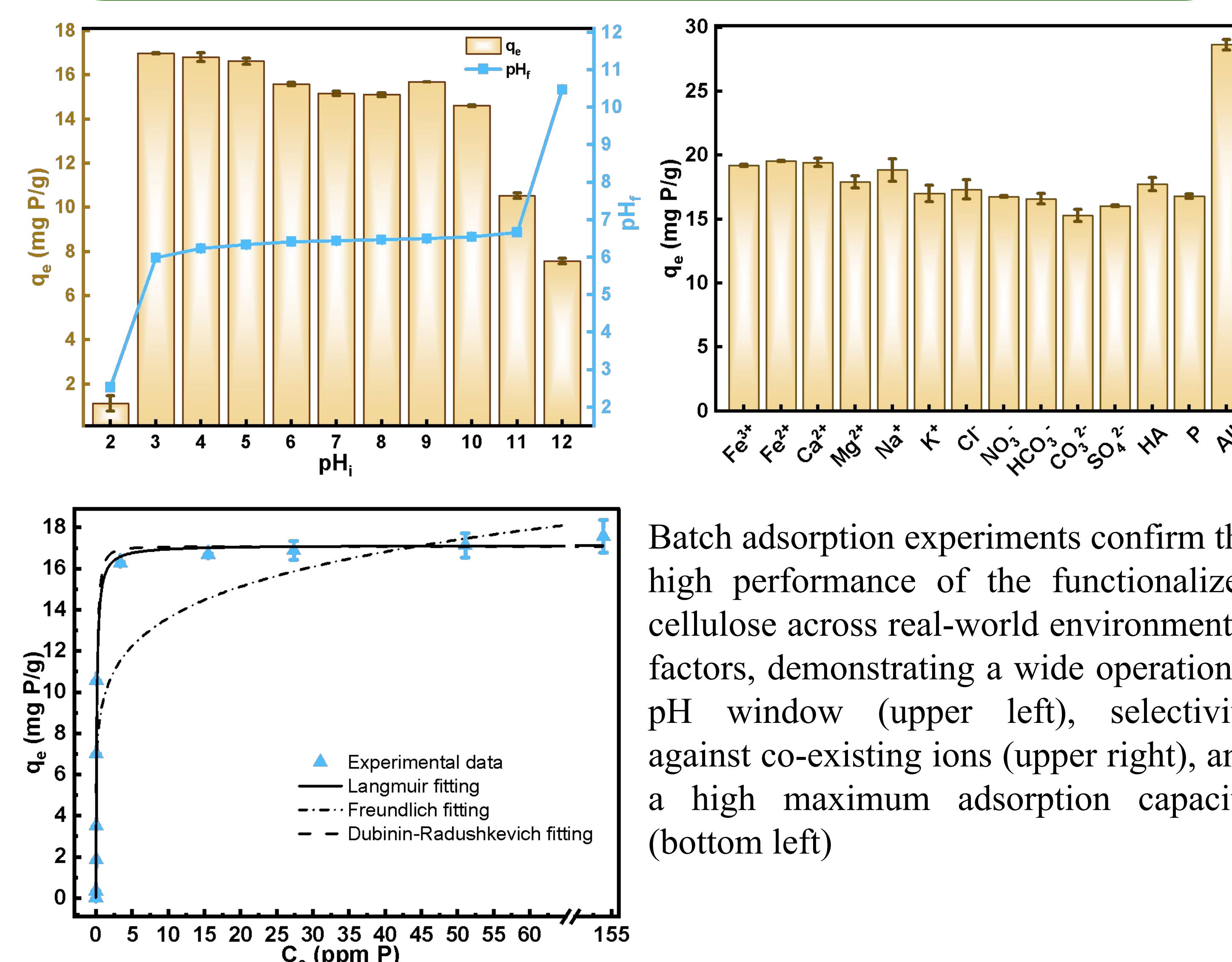


Basil

- ### Phosphorus Uptake
- Measure P in plant tissue



From Lab to Market: Economic Feasibility



Batch adsorption experiments confirm the high performance of the functionalized cellulose across real-world environmental factors, demonstrating a wide operational pH window (upper left), selectivity against co-existing ions (upper right), and a high maximum adsorption capacity (bottom left)

Loop Restored: Conclusion & Future Outlook

- ❖ Sustainable Feedstock Upcycling: Successfully extracted cellulose from waste paper to divert high-value biomass from landfills, confirmed by EDX/SEM
- ❖ Phosphorus Nutrient Capture: Validated the functionalization of active cerium nanoparticles and successfully capture the phosphate coordination
- ❖ Closed-Loop Resource Restoration: Stress-tested the adsorbent across crucial real-world environmental factors and effectively repairing the broken phosphorus cycle to prevent resource depletion and environmental damage

Acknowledgements



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